Marco Borriero

PhD Candidate in Statistics

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RESEARCH PROFILE

My research interests focus on the theoretical foundations and methodological advances of causal inference, causal discovery, graphical models, and functional data analysis. In particular, I work on establishing new statistical frameworks that leverage the causal invariance assumption to achieve simultaneous identification of direct causes for a given target variable and causal effect estimation. I am broadly motivated by translating complex real-world problems into rigorous theoretical advances.

CURRENT POSITION

PhD Candidate in Statistics at University of Florence Nov 2023 - Present
Department of Statistics, Computer Science, Applications, Florence, Italy
Supervisors: Prof. Monia Lupparelli and Prof. Kayvan Sadeghi
Thesis: Causal Invariance in Graphical Models with Complex Structures
Thesis status: Submitted
Expected graduation: February/March 2027

EDUCATION

Master's Degree in Mathematics and Statistics Sept 2021 - Oct 2023
University of Trento, Italy
Graduated with 110/110 (cum laude)

Bachelor's Degree in Mathematics Sept 2018 - Sept 2021
University of Trento, Italy

VISITING POSITION

Visiting PhD student at University College London Jan 2026 - Mar 2026
Department of Statistical Science, London, United Kingdom
Supervisor: Prof. Kayvan Sadeghi
Focus: Preliminary results for the definition of a theory of causality starting from soft interventional distribution probabilities

TEACHING EXPERIENCE

Tutor for Bachelor's Calculus Course, University of Florence, Feb 2024 - May 2024
Department of Economics

- Led weekly tutorial sessions for undergraduate students, solving practical exercises
- Provided one-to-one student support during office hours

PUBLICATIONS

Journal paper

· Borriero, M., Augugliaro, L., Sottile, G., & Vinciotti, V. (2026). Gaussian Graphical Models for Partially Observed Multivariate Functional Data. *Journal of Computational and Graphical Statistics*. doi: `https://doi.org/10.1080/10618600.2026.2756975`. *[In press]*

Preprint

· Borriero M., Lupparelli M., Marchetti G. M., Vinciotti V. (2026). Causal invariance in graphical models with latent variables. `https://arxiv.org/abs/2606.13281`.

Conference papers

· Borriero M., Lupparelli M., Marchetti G. M., Vinciotti, V. (2025). Invariant Causal Prediction for Gaussian Multivariate Regression Graphs. In *Scientific Meeting of the Italian Statistical Society* (pp. 213-219). Cham: Springer Nature Switzerland.

· Borriero M., Augugliaro L., Latora S., Vinciotti V. (2024). Completion of Partially Observed Multivariate Functional Data. In *Scientific Meeting of the Italian Statistical Society* (pp. 254-259). Cham: Springer Nature Switzerland.

INTERNATIONAL CONFERENCE PRESENTATIONS

Invited

· Causal invariance in graphical models with latent variables, Data Science, Statistics and Visualization (DSSV 2026), Trento, Italy, June 2026

Contributed

· Invariant causal prediction under causal insufficiency, 27th International Conference on Computational Statistics (COMPSTAT 2026), Athens, Greece, August 2026

· Testing causal invariance for a multivariate target under causal insufficiency, European Meeting of Statisticians (EMS 2026, Bernoulli Society), Lugano, Switzerland, August 2026

· Invariant causal prediction for Gaussian multivariate regression graphs, Statistics for Innovation, Genoa, Italy, June 2025

· Completion of partially observed multivariate functional data, The 52nd Scientific Meeting of the Italian Statistical Society, Bari, Italy, June 2024

PROFESSIONAL SERVICES

Reviewer for the **Scandinavian Journal of Statistics**

SKILLS

Programming: R, Python, MATLAB, JavaScript, LaTeX

Languages: Italian (Native speaker), English (Fluent), French (Beginner)